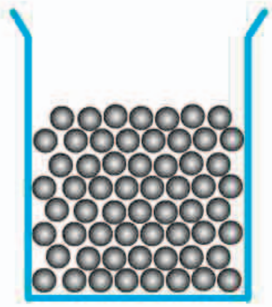
Liquids

In a liquid, most of the atoms and molecules are still bonded together, but about one in ten of the links between them is broken. As a result, a liquid still has a more or less fixed volume and density—squeezing it does not really reduce its volume much. However, the constituents of the liquid are freer to move around than in a solid. Liquid can flow down slopes under the force of gravity, and take on the shape of any container it is poured into.



◁ **Liquid metal**
Mercury is the only metal that is liquid at room temperature. This is because its atoms form only weak bonds with each other.

honey is very viscous and flows slowly

oil is less viscous, but does not splash much

water has low viscosity, and drips and splashes easily

◁ **Viscosity**
How a liquid flows is called its viscosity. When molecules are often blocked from moving past each other, the liquid is viscous (thick) and flows slowly. In low-viscosity liquids, molecules move around with little resistance.

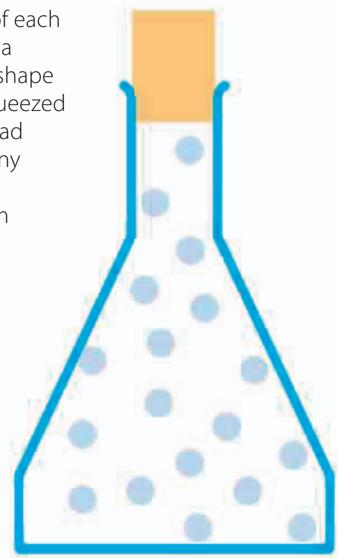
REAL WORLD

Plasma

The aurora, or Northern Lights and Southern Lights, are an example of a fourth state of matter: plasma. Plasma is a mixture of high-energy charged atoms and smaller subatomic particles. The aurora is formed by plasma streaming from the Sun being trapped in the Earth's magnetic field. It crashes into the atmosphere over the polar regions, creating the amazing light show.

Gases

In a gas, there are no bonds at all between the atoms or molecules. The units are free to move independently of each other in any direction. As a result, a gas has no fixed shape or volume and can be squeezed into a small space or spread out to fill a container of any shape. Like a liquid, it can also be made to flow from one place to another.



▷ **Helium**
Helium is made up of just single atoms. As they move around, the atoms bounce off each other and the sides of the container.